interface.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/AsyncFifo/interface) - GVIM

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```
interface fifo_if #(
        parameter DATA_WIDTH    =32,
        parameter FIFO_DEPTH    =16)

        ();
        logic wclk;
        logic rclk;

        logic                    wrst_n;
        logic                    wr_en;
        logic [DATA_WIDTH-1:0]   wr_data;
        logic                    wfull;
        logic                    walmost_full;
        logic [$clog2(FIFO_DEPTH):0] wfill_count;
        logic                    overflow_error;
        logic                   clr_overflow;


        logic                   rrst_n;
        logic                   rd_en;
        logic [DATA_WIDTH-1:0]  rd_data;
        logic                   rempty;
        logic                   ralmost_empty;
        logic [$clog2(FIFO_DEPTH):0] rfill_count;
        logic                   underflow_error;
        logic                   clr_underflow;
        logic [4:0]             wgray;
        logic [4:0]             rgray;

        int wclk_period=10;
        int rclk_period=10;
        int rclk_phase=0;

        task run_wclk();
                wclk=0;
                forever #(wclk_period/2) wclk=~wclk;
        endtask

        task run_rclk();
                rclk=0;
```

```
int rclk_period=10;
int rclk_phase=0;

task run_wclk();
        wclk=0;
        forever #(wclk_period/2) wclk=~wclk;
endtask

task run_rclk();
        rclk=0;
#(rclk_phase);
forever #(rclk_period/2) rclk=~rclk;
endtask

clocking wdrv_cb @(posedge wclk);
        input wfull;
        input walmost_full;
        input wfill_count;
        input overflow_error;
        input wgray;

        output clr_overflow;
        output wrst_n;
        output wr_en;
        output wr_data;
endclocking

clocking rdrv_cb @(posedge rclk);
        input rempty;
        input ralmost_empty;
        input rfill_count;
        input underflow_error;
        input rd_data;
        input rgray;

        output clr_underflow;
        output rrst_n;
        output rd_en;
endclocking
```

```
            input rgray;

            output clr_underflow;
            output rrst_n;
            output rd_en;
        endclocking

        clocking wmon_cb @(posedge wclk);
            input wfull;
            input walmost_full;
            input wfill_count;
            input overflow_error;
            input wrst_n;
            input wr_en;
            input wr_data;
            input clr_overflow;
            input wgray;
        endclocking

        clocking rmon_cb @(posedge rclk);
            input rempty;
            input ralmost_empty;
            input rfill_count;
            input underflow_error;
            input rrst_n;
            input rd_en;
            input rd_data;
            input clr_underflow;
            input rgray;
        endclocking


        modport wdrv(clocking wdrv_cb);
        modport rdrv(clocking rdrv_cb);
        modport wmon(clocking wmon_cb);
        modport rmon(clocking rmon_cb);

endinterface
```

```
`timescale 1ns/1ps

module async_fifo_top;

        `include "uvm_macros.svh"
        import uvm_pkg::*;

        import fifo_pkg::*;

        logic wclk,rclk;
        logic wrst_n,rrst_n;

        fifo_if #(.DATA_WIDTH(32),
                .FIFO_DEPTH(16))vif();

        initial begin
                vif.run_wclk();
        end
        initial begin
                vif.run_rclk();
        end

        async_fifo  #(.DATA_WIDTH(32),.FIFO_DEPTH(16)) dut (
                .wclk(vif.wclk),
                .wrst_n(vif.wrst_n),
                .wr_en(vif.wr_en),
                .wr_data(vif.wr_data),
                .wfull(vif.wfull),
                .walmost_full(vif.walmost_full),
                .wfill_count(vif.wfill_count),
                .overflow_error(vif.overflow_error),
                .clr_overflow(vif.clr_overflow),


                .rclk(vif.rclk),
                .rrst_n(vif.rrst_n),
                .rd_en(vif.rd_en),
                .rd_data(vif.rd_data),
                .rempty(vif.rempty),
                .ralmost_empty(vif.ralmost_empty),
```

```
                .wr_data(vif.wr_data),
                .wfull(vif.wfull),
                .walmost_full(vif.walmost_full),
                .wfill_count(vif.wfill_count),
                .overflow_error(vif.overflow_error),
                .clr_overflow(vif.clr_overflow),


                .rclk(vif.rclk),
                .rrst_n(vif.rrst_n),
                .rd_en(vif.rd_en),
                .rd_data(vif.rd_data),
                .rempty(vif.rempty),
                .ralmost_empty(vif.ralmost_empty),
                .rfill_count(vif.rfill_count),
                .underflow_error(vif.underflow_error),
                .clr_underflow(vif.clr_underflow)
                        );

        bind async_fifo gray_bind bind_inst(.wclk(wclk),.rclk(rclk),.wgray(wgray),.rgray(rgray));
        bind async_fifo async_fifo_sva #(.DEPTH(16),.PTR_WIDTH(5))fifo_assert_inst(.waddr(waddr),.wgray(wgray),.rgray(rgray),.raddr(raddr),
                .wclk(wclk),.rclk(rclk),.wrst_n(wrst_n),.rrst_n(rrst_n),.wr_en(wr_en),.rd_en(rd_en),.wr_data(wr_data),.rd_data(rd_data),.wfull(wfull),.rempty(rempty),.wgray_sync(wgray_sync)
);

        initial begin

                uvm_config_db#(virtual fifo_if)::set(null,"*","vif",vif);
                uvm_top.print_config();
                run_test();
        end

                initial begin
                        #200000;
                        `uvm_fatal("TIMEOUT","simulation timeout reached!")
                end
endmodule
```

```
class async_base_test extends uvm_test;
        async_fifo_env env;
        virtual fifo_if vif;
        `uvm_component_utils(async_base_test)

        function new(string name="async_base_test",uvm_component parent);
                super.new(name,parent);
        endfunction

        virtual function void build_phase(uvm_phase phase);
                super.build_phase(phase);

                if(!uvm_config_db #(virtual fifo_if)::get(this,"","vif",vif)) begin
                        `uvm_fatal("test","virtual interface not set")
                end
                env=async_fifo_env::type_id::create("env",this);

        endfunction
        virtual function void end_of_elaboration_phase(uvm_phase phase);
                super.end_of_elaboration_phase(phase);

                uvm_top.print_topology();
        endfunction

        task reset_phase(uvm_phase phase);
                phase.raise_objection(this);
                `uvm_info(get_type_name(),"applying write reset",UVM_MEDIUM)
                apply_wreset();
                `uvm_info(get_type_name(),"applying read reset",UVM_MEDIUM)
                apply_rreset();
                `uvm_info(get_type_name(),"reset complete",UVM_MEDIUM)
                phase.drop_objection(this);
        endtask

        virtual task apply_wreset();
                virtual fifo_if vif;
                if(!uvm_config_db#(virtual fifo_if)::get(this,"","vif",vif))
                        `uvm_fatal("NOVIF","virtual interface not found in base test")

                vif.wrst_n<=0;
```

```
		`uvm_info(get_type_name(),"applying read reset",UVM_MEDIUM)
		apply_rreset();
		`uvm_info(get_type_name(),"reset complete",UVM_MEDIUM)
		phase.drop_objection(this);
	endtask

	virtual task apply_wreset();
		virtual fifo_if vif;
		if(!uvm_config_db#(virtual fifo_if)::get(this,"","vif",vif))
			`uvm_fatal("NOVIF","virtual interface not found in base test")

		vif.wrst_n<=0;
		repeat(5) @(posedge vif.wclk);
		vif.wrst_n<=1;
		repeat(2) @(posedge vif.wclk);
	endtask

	virtual task apply_rreset();
		virtual fifo_if vif;
		if(!uvm_config_db#(virtual fifo_if)::get(this,"","vif",vif))
			`uvm_fatal("NOVIF","virtual interface not found in base test")

		vif.rrst_n<=0;
		repeat(5) @(posedge vif.rclk);
		vif.rrst_n<=1;
		repeat(2) @(posedge vif.rclk);
	endtask

	virtual function void report_phase(uvm_phase phase);
		uvm_report_server rs=uvm_report_server::get_server();
		int error_count=rs.get_severity_count(UVM_ERROR);
		int fatal_count=rs.get_severity_count(UVM_FATAL);

		if(error_count>0 || fatal_count>0)
			`uvm_info(get_type_name(),"\n test failed",UVM_NONE)
		else
			`uvm_info(get_type_name(),"\n test passed",UVM_NONE)
	endfunction
endclass
```

```
class async_fifo_env extends uvm_env;

        virtual fifo_if vif;

        `uvm_component_utils(async_fifo_env)

        write_agent wr_ag;
        read_agent rd_ag;
        scoreboard sb;
        coverage cg;

        function new(string name="async_fifo_env",uvm_component parent);
                super.new(name,parent);
        endfunction

        function void build_phase(uvm_phase phase);
                super.build_phase(phase);
                wr_ag=write_agent::type_id::create("wr_ag",this);
                rd_ag=read_agent::type_id::create("rd_ag",this);
                sb=scoreboard::type_id::create("sb",this);
                cg=coverage::type_id::create("cg",this);
                uvm_config_db#(virtual fifo_if)::set(this,"wr_ag.*","vif",vif);
                uvm_config_db#(virtual fifo_if)::set(this,"rd_ag.*","vif",vif);
        endfunction

        function void connect_phase(uvm_phase phase);
                super.connect_phase(phase);
                wr_ag.wr_mon.ap.connect(sb.wr_imp);
                rd_ag.rd_mon.ap.connect(sb.rd_imp);
                wr_ag.wr_mon.ap.connect(cg.wr_imp);
                rd_ag.rd_mon.ap.connect(cg.rd_imp);


        endfunction
endclass
```

```
package fifo_pkg;
        `include "uvm_macros.svh"

        import uvm_pkg ::*;


        `include "../agents/transaction.sv"
        `include "../agents/write_driver.sv"
        `include "../agents/wr_sequencer.sv"
        `include "../agents/wr_monitor.sv"
        `include "../agents/wr_agent.sv"


        `include "../agents/rd_driver.sv"
        `include "../agents/rd_monitor.sv"
        `include "../agents/rd_sequencer.sv"
        `include "../agents/rd_agent.sv"


        `include "scoreboard.sv"
        `include "coverage.sv"
        `include "async_fifo_env.sv"

        `include "../sequences/async_seq_lib.sv"

        `include "../tests/asyn_fifo_tests.sv"

endpackage
~
~
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~
~
```

transaction.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/AsyncFifo/agent

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rd_agent.sv rd_driver.sv rd_monitor.sv rd_sequencer.sv **transaction.sv** wr_agent.sv write_driver.sv wr_monitor.sv wr_sequencer.sv

```
class fifo_transaction extends uvm_sequence_item;
        rand bit wrst_n;
        rand bit rrst_n;
        rand bit [31:0] wdata;
        rand bit wr_en;
        rand bit rd_en;
        bit [31:0] rdata;
         rand bit clr_overflow;
         rand bit clr_underflow;
        bit rempty;
        bit wfull;
        bit almostfull;
        bit almostempty;
        rand bit clk_ratio;
         int wgray;
         int rgray;
        bit [$clog2(16):0] wfill_count;
        bit [$clog2(16):0]rfill_count;
        bit overflow_error;
        bit underflow_error;
        rand int wclk_period;
        rand int rclk_period;
        int ratio_scaled;
        rand int write_count;
        rand int read_count;
        rand int unsigned fifo_level;

        parameter DEPTH=16;


        `uvm_object_utils_begin(fifo_transaction)
                `uvm_field_int(wdata,UVM_ALL_ON)
                `uvm_field_int(wr_en,UVM_ALL_ON)
                `uvm_field_int(rd_en,UVM_ALL_ON)
                `uvm_field_int(rdata,UVM_ALL_ON)
                `uvm_field_int(wrst_n,UVM_ALL_ON)
                `uvm_field_int(rrst_n,UVM_ALL_ON)
                `uvm_field_int(clr_overflow,UVM_ALL_ON)
                `uvm_field_int(clr_underflow,UVM_ALL_ON)
```

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rd_agent.sv | rd_driver.sv | rd_monitor.sv | rd_sequencer.sv | transaction.sv | wr_agent.sv | write_driver.sv | wr_monitor.sv | wr_sequencer.sv

```
        `uvm_field_int(rrst_n,UVM_ALL_ON)
        `uvm_field_int(clr_overflow,UVM_ALL_ON)
        `uvm_field_int(clr_underflow,UVM_ALL_ON)
        `uvm_field_int(rempty,UVM_ALL_ON)
        `uvm_field_int(wfull,UVM_ALL_ON)
        `uvm_field_int(almostfull,UVM_ALL_ON)
        `uvm_field_int(almostempty,UVM_ALL_ON)
        `uvm_field_int(clk_ratio,UVM_ALL_ON)
        `uvm_field_int(wfill_count,UVM_ALL_ON)
        `uvm_field_int(wgray,UVM_ALL_ON)
        `uvm_field_int(rgray,UVM_ALL_ON)
        `uvm_field_int(ratio_scaled,UVM_ALL_ON)
        `uvm_field_int(overflow_error,UVM_ALL_ON)
        `uvm_field_int(underflow_error,UVM_ALL_ON)
        `uvm_field_int(wclk_period,UVM_ALL_ON)
        `uvm_field_int(rclk_period,UVM_ALL_ON)
        `uvm_field_int(write_count,UVM_ALL_ON)
        `uvm_field_int(read_count,UVM_ALL_ON)
        `uvm_field_int(fifo_level,UVM_ALL_ON)
    `uvm_object_utils_end

    constraint wr_c{
        wr_en dist {1:=50 , 0:=50};}
    constraint rd_c{
        rd_en dist {1:=50 , 0:=50};}
    constraint rst_c{
        wrst_n dist {1:=5, 0:=95};
        rrst_n dist {1:=5, 0:=95};
    }

    constraint data_range_c {
        wdata inside {[32'h0000_0000:32'hFFFF_FFFF]};
    }
    constraint w_data_c {
        wdata dist{
        32'h0000_0000                   :/5,
        [32'h0000_0001:32'h0000_FFFF]   :/5,
        [32'h0001_0000:32'h7FFF_FFFF]   :/90,
        [32'h8000_0000:32'hFFFF_FFFF]   :/90
```

-- INSERT --

transaction.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/Un

File Edit Tools Syntax Buffers Window Help

rd_agent.sv | rd_driver.sv | rd_monitor.sv | rd_sequencer.sv | transaction.sv | wr_agent.sv | write_driver.sv | wr_monitor.sv

```
    constraint w_data_c {
            wdata dist{
            32'h0000_0000                   :/5,
            [32'h0000_0001:32'h0000_FFFF]   :/5,
            [32'h0001_0000:32'h7FFF_FFFF]   :/90,
            [32'h8000_0000:32'hFFFF_FFFF]   :/90

    };}

    constraint valid_counts_c{
            write_count inside {[0:15]};    //no underflow
            read_count inside {[0:15]};     //no overflow
    }

  constraint write_count_c{
            write_count inside {[0:DEPTH]};
            write_count dist {
                 [0:5] :=10,
                 [6:10] :=80,
                 [11:15] :=70
            };
    }

    constraint read_count_c{
            read_count inside {[0:DEPTH]};
            read_count dist{
                 [0:5] :=5,
                 [6:10] :=70,
                 [11:15] :=80
            };
    }
  constraint rst_cross_c{
            (wrst_n==0 && rrst_n==0) ||
            (wrst_n==0 && rrst_n==1) ||
            (wrst_n==1 && rrst_n==0) ||
            (wrst_n==1 && rrst_n==1);
  }

    constraint fifo_level_c{
```

-- INSERT --

transaction.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVe

File Edit Tools Syntax Buffers Window Help

rd_agent.sv | rd_driver.sv | rd_monitor.sv | rd_sequencer.sv | transaction.sv | wr_agent.sv | write_driver.sv | wr_monitor.sv | wr_s

```
                [6:10] :=70,
                [11:15] :=80
            };
      }
    constraint rst_cross_c{
            (wrst_n==0 && rrst_n==0) ||
            (wrst_n==0 && rrst_n==1) ||
            (wrst_n==1 && rrst_n==0) ||
            (wrst_n==1 && rrst_n==1);
    }

       constraint fifo_level_c{
              fifo_level dist{
                     0       :=1,
                     [1:5]   :=3,
                     [6:10]  :=4,
                     [11:14] :=4,
                     15      :=2
       };}

       function void post_randomize();
       ratio_scaled=(wclk_period*100)/rclk_period;
endfunction

       function new(string name="fifo_transaction");
              super.new(name);
       endfunction
endclass

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```

rd_agent.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/AsyncFifo/agents) (1 of 9) - GVIM5 (

File Edit Tools Syntax Buffers Window Help

rd_agent.sv rd_driver.sv rd_monitor.sv rd_sequencer.sv transaction.sv wr_agent.sv write_driver.sv wr_monitor.sv wr_sequencer.sv

```
class read_agent extends uvm_agent;
        `uvm_component_utils(read_agent)

        uvm_sequencer #(fifo_transaction) rd_sqr;
        read_driver rd_drv;
        read_monitor rd_mon;

        function new(string name="read_agent",uvm_component parent);
                super.new(name,parent);
        endfunction

        function void build_phase(uvm_phase phase);
                super.build_phase(phase);
                rd_mon=read_monitor::type_id::create("rd_mon",this);
                if(get_is_active()==UVM_ACTIVE) begin
                        rd_sqr=read_sequencer::type_id::create("rd_sqr",this);
                        rd_drv=read_driver::type_id::create("rd_drv",this);
                end
        endfunction

        function void connect_phase(uvm_phase phase);
                if(is_active==UVM_ACTIVE)
                        rd_drv.seq_item_port.connect(rd_sqr.seq_item_export);
        endfunction
endclass

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```

```
class read_driver extends uvm_driver#(fifo_transaction);
        `uvm_component_utils(read_driver)
        virtual fifo_if vif;
        int read_count;

        function new(string name="read_driver",uvm_component parent);
                super.new(name,parent);
        endfunction

        function void build_phase(uvm_phase phase);
                super.build_phase(phase);
                if(!uvm_config_db #(virtual fifo_if)::get(this,"","vif",vif))
                        `uvm_fatal("drv","no virtual interface  set for read driver")
        endfunction

        task run_phase(uvm_phase phase);

                fifo_transaction tx;
                read_count=0;
                vif.rdrv_cb.rd_en<=0;
                vif.rdrv_cb.clr_underflow<=0;
                forever begin
                        seq_item_port.get_next_item(tx);
                        `uvm_info("drv","driver running",UVM_LOW)
                        @(vif.rdrv_cb);
                        if(tx.rd_en && !(vif.rdrv_cb.rempty)) begin
                //      @(vif.rdrv_cb);
                                vif.rdrv_cb.rd_en<=1;
                                vif.rdrv_cb.clr_underflow<=tx.clr_underflow;
                                `uvm_info("drv",$sformatf("rd_en=%0b ,clr_underflow=%0b",tx.rd_en,tx.clr_underflow),UVM_LOW)
                        end
                        else begin
                        @(vif.rdrv_cb);
                                vif.rdrv_cb.rd_en<=0;
                        seq_item_port.item_done();
                        read_count++;
                        `uvm_info("rdrv",$sformatf("read count=%0d",read_count),UVM_LOW)
                //      @(vif.rdrv_cb);
                        end
                        end
        endtask
endclass
```

rd_monitor.sv + (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/AsyncFifo/agents) (3 of 9) - GVIM5 (on ip-10-115-24-57.cadence.com)

File Edit Tools Syntax Buffers Window Help

rd_agent.sv rd_driver.sv + rd_monitor.sv rd_sequencer.sv transaction.sv wr_agent.sv write_driver.sv wr_monitor.sv wr_sequencer.sv

```
class read_monitor extends uvm_monitor;
        `uvm_component_utils(read_monitor)
        virtual fifo_if vif;
        int read_count;
        int fifo_level;
        uvm_analysis_port #(fifo_transaction) ap;
        function new(string name="read_monitor",uvm_component parent);
                super.new(name,parent);
                ap=new("ap",this);
        endfunction
        function void build_phase(uvm_phase phase);
                super.build_phase(phase);
                if(!uvm_config_db #(virtual fifo_if)::get(this,"","vif",vif)) begin
                        `uvm_fatal("rmon","virtual interface not set")
                end
        endfunction
        task run_phase(uvm_phase phase);
                fifo_transaction tx;
                read_count=0;
                forever begin
          while(vif.rmon_cb.rd_en==1'b0 || vif.rmon_cb.rempty==1'b1)
                                @(vif.rmon_cb);
          if(vif.rmon_cb.rd_en==1 && vif.rmon_cb.rempty==1'b0)begin:if_block
                                @(vif.rmon_cb);
          `uvm_info("READ_CP","at this time it will capture a read packet",UVM_LOW)
                                tx=fifo_transaction::type_id::create("tx",this);
                                tx.rrst_n=vif.rmon_cb.rrst_n;
                                tx.rd_en=vif.rmon_cb.rd_en;
                                tx.rdata=vif.rmon_cb.rd_data;
                                tx.rempty=vif.rmon_cb.rempty;
                                tx.clr_underflow=vif.rmon_cb.clr_underflow;
                                tx.rempty=vif.rmon_cb.rempty;
                                tx.almostempty=vif.rmon_cb.ralmost_empty;
                                tx.rfill_count=vif.rmon_cb.rfill_count;
                                tx.underflow_error=vif.rmon_cb.underflow_error;
                                tx.fifo_level=fifo_level;
                                ap.write(tx);
                                read_count++;
                                fifo_level--;
                                `uvm_info("READ_SEND",$sformatf("$time=%0t\trd_en=%0b  and rdata=%0h rempty=%0h,read count=%0d,fifo level=%0d",$time,vif.rmon_cb.rd_en,vif.rmon_cb.rd_data,vif.rmon_cb.rempty,read
_count,fifo_level),UVM_MEDIUM)
                        //      `uvm_info("rd mon","sending transaction to scoreboard",UVM_LOW)
          end:if_block
          else
                                @(vif.rmon_cb);

                end
        endtask
```

-- INSERT -- 6,43-50 Top

Mate Terminal | shyd2508dv002@ip-10-... | fifo_pkg.sv (/projects/a... | fifo_mem.sv (/projects/... | interface.sv (/projects/a... | top_tb.sv (/projects/am... | asyn_fifo_tests.sv (/proj... | rd_monitor.sv + (/projec...

```
class write_agent extends uvm_agent;
        `uvm_component_utils(write_agent)

        uvm_sequencer #(fifo_transaction) wr_sqr;
        write_driver wr_drv;
        write_monitor wr_mon;

        function new(string name="write_agent",uvm_component parent);
                super.new(name,parent);
        endfunction

        function void build_phase(uvm_phase phase);
                super.build_phase(phase);
                wr_mon=write_monitor::type_id::create("wr_mon",this);
                if(get_is_active()== UVM_ACTIVE) begin
                wr_sqr=write_sequencer::type_id::create("wr_sqr",this);
                wr_drv=write_driver::type_id::create("wr_drv",this);
        end
        endfunction

        function void connect_phase(uvm_phase phase);
                if(get_is_active()==UVM_ACTIVE)
                        wr_drv.seq_item_port.connect(wr_sqr.seq_item_export);
        endfunction
endclass


~
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```

write_driver.sv (/projects/am61_proj_dv/shyd2508dv002/sumedha/ActiveProjects/AsyncFifo/10_geethika/verification/UnitVerif/AsyncFifo/agents) (7 of 9) - GVIM5 (on ip-10-115-24-57.cadence.com)

File Edit Tools Syntax Buffers Window Help

rd_agent.sv | rd_driver.sv | rd_monitor.sv | rd_sequencer.sv | transaction.sv | wr_agent.sv | write_driver.sv | wr_monitor.sv | wr_sequencer.sv

```
class write_driver extends uvm_driver #(fifo_transaction);
        `uvm_component_utils(write_driver)
        virtual fifo_if vif;
        int write_count;

        function new(string name="write_driver",uvm_component parent);
                super.new(name,parent);
        endfunction

        function void build_phase(uvm_phase phase);
                super.build_phase(phase);
                if(!uvm_config_db#(virtual fifo_if)::get(this,"","vif",vif))
                        `uvm_fatal("drv","no virtual interface set for write driver")

        endfunction

        task run_phase(uvm_phase phase);

               fifo_transaction tx;
               vif.wdrv_cb.wr_en<=0;
               vif.wdrv_cb.wr_data<=32'h0;
               vif.wdrv_cb.clr_overflow<=0;
               write_count=0;
                forever begin
                        seq_item_port.get_next_item(tx);
                        `uvm_info("drv","driver running",UVM_LOW)
                        if(tx.wr_en && !tx.wfull) begin
                                @(vif.wdrv_cb);
                                        vif.wdrv_cb.wr_en<=1'd1;
                                        vif.wdrv_cb.wr_data<=tx.wdata;
                                        vif.wdrv_cb.clr_overflow<=tx.clr_overflow;
                                        `uvm_info("drv",$sformatf("wr_en=%0b ,clr_overflow=%0b wr data=%0h",tx.wr_en,tx.clr_overflow,tx.wdata),UVM_MEDIUM)
                                @(vif.wdrv_cb);
                                        vif.wdrv_cb.wr_en<=0;

                        end
                        seq_item_port.item_done();
                        write_count++;
                        `uvm_info("wdrv",$sformatf("write count=%0d",write_count),UVM_LOW)

                end
        endtask
endclass
```

-- INSERT -- 1,1

```
class write_monitor extends uvm_monitor;
        `uvm_component_utils(write_monitor)
        virtual fifo_if vif;
        int write_count;
        int fifo_level;
        uvm_analysis_port #(fifo_transaction) ap;
        function new(string name="write_monitor",uvm_component parent);
                super.new(name,parent);
                ap=new("ap",this);
        endfunction

        function void build_phase(uvm_phase phase);
                super.build_phase(phase);
                if(!uvm_config_db#(virtual fifo_if)::get(this,"","vif",vif)) begin
                        `uvm_fatal("wmon","virtual interface not set");
                end
        endfunction

        task run_phase(uvm_phase phase);
                        fifo_transaction tx;
                        write_count=0;
                        fifo_level=0;
                forever begin
                        if(vif.wmon_cb.wr_en && !(vif.wmon_cb.wfull)) begin
                        @(vif.wmon_cb);
              `uvm_info("WRITE_CP","at this time it will capture a write packet",UVM_LOW)
                         tx=fifo_transaction::type_id::create("tx",this);
                         tx.wrst_n=vif.wmon_cb.wrst_n;
                        tx.wr_en=1;
                         tx.wdata=vif.wmon_cb.wr_data;
                         tx.wfull=vif.wmon_cb.wfull;
                         tx.clr_overflow=vif.wmon_cb.clr_overflow;
                         tx.almostfull=vif.wmon_cb.walmost_full;
                         tx.wfill_count=vif.wmon_cb.wfill_count;
                         tx.overflow_error=vif.wmon_cb.overflow_error;
                         tx.fifo_level=fifo_level;
                         ap.write(tx);
                         write_count++;
                         fifo_level++;
                         `uvm_info("WRITE_SEND",$sformatf(" wr_en=%0b ,wr data=%0h,write count=%0d,fifo level=%0d",vif.wmon_cb.wr_en,vif.wmon_cb.wr_data,write_count,fifo_level),UVM_LOW)
                        end
                 else
                        @(vif.wmon_cb);
                end
        endtask
```

```
`uvm_analysis_imp_decl(_wr)
`uvm_analysis_imp_decl(_rd)
class scoreboard extends uvm_scoreboard;
        `uvm_component_utils(scoreboard)

        uvm_analysis_imp_wr #(fifo_transaction,scoreboard) wr_imp;
        uvm_analysis_imp_rd #(fifo_transaction,scoreboard) rd_imp;

        bit [31:0] exp_q[$];
        bit [31:0] req_q;

        parameter DATA_WIDTH=32;
        parameter FIFO_DEPTH=16;

        int match_count;
        int mismatch_count;
        int total_writes;
        int total_reads;
        int total_valid_reads;
        int total_compares;

        function new(string name="scoreboard",uvm_component parent);
                super.new(name,parent);
                wr_imp=new("wr_imp",this);
                rd_imp=new("rd_imp",this);
        //      total_writes=0;
        //      total_reads=0;
        //      total_valid_reads=0;
        //      match_count=0;
        //      mismatch_count=0;
        endfunction

        function void build_phase(uvm_phase phase);
                super.build_phase(phase);
                reset();
        endfunction

        function void write_wr(fifo_transaction tx);
              `uvm_info("SB_WRITE_CP","at this time it will capture a write packet",UVM_LOW)
-- INSERT --
```